

Appendix I: Conversions

Appendix I: Unit Conversions and Constants

I.1 Length Conversions

From/To	Millimeter (mm)	Centimeter (cm)	Meter (m)	Inch (in)	Foot (ft)
1 mm	1	0.1	0.001	0.03937	0.003281
1 cm	10	1	0.01	0.3937	0.03281
1 m	1000	100	1	39.37	3.281
1 in	25.4	2.54	0.0254	1	0.08333
1 ft	304.8	30.48	0.3048	12	1

Common CNC Conversions: - 1 inch = 25.4 mm (exact) - 1 foot = 304.8 mm - 1 mil (0.001") = 0.0254 mm = 25.4 μ m - 1 micron (μ m) = 0.001 mm = 0.00003937"

I.2 Area and Volume Conversions

I.2.1 Area

From/To	mm ²	cm ²	m ²	in ²	ft ²
1 mm²	1	0.01	10 ⁻⁶	0.00155	1.076x10 ⁻⁵
1 cm²	100	1	10 ⁻⁴	0.155	0.001076
1 m²	10 ⁶	10 ⁴	1	1550	10.76
1 in²	645.2	6.452	6.452x10 ⁻⁴	1	0.00694
1 ft²	92,903	929.03	0.0929	144	1

I.2.2 Volume

From/To	cm ³ (cc)	Liter (L)	m ³	in ³	ft ³	US Gallon
1 cm³	1	0.001	10 ⁻⁶	0.06102	3.531x10 ⁻⁵	0.000264
1 L	1000	1	0.001	61.02	0.03531	0.2642
1 m³	10 ⁶	1000	1	61,024	35.31	264.2
1 in³	16.39	0.01639	1.639x10 ⁻⁵	1	5.787x10 ⁻⁴	0.004329
1 ft³	28,317	28.32	0.02832	1728	1	7.481
1 US gal	3785	3.785	0.003785	231	0.1337	1

I.3 Force, Pressure, and Torque Conversions

I.3.1 Force

From/To	Newton (N)	Kilonewton (kN)	Kilogram-force (kgf)	Pound-force (lbf)
1 N	1	0.001	0.102	0.2248
1 kN	1000	1	102.0	224.8
1 kgf	9.807	0.009807	1	2.205
1 lbf	4.448	0.004448	0.4536	1

I.3.2 Pressure and Stress

From/To	Pascal (Pa)	kPa	MPa	Bar	PSI	kg/cm ²
1 Pa	1	0.001	10 ⁻⁶	10 ⁻⁵	1.450x10 ⁻⁴	1.020x10 ⁻⁵
1 kPa	1000	1	0.001	0.01	0.1450	0.01020
1 MPa	10 ⁶	1000	1	10	145.0	10.20
1 bar	10 ⁵	100	0.1	1	14.50	1.020
1 PSI	6895	6.895	0.006895	0.06895	1	0.07031
1 kg/cm²	98,067	98.07	0.09807	0.9807	14.22	1

Common Pneumatic Pressures: - 100 PSI = 6.9 bar = 0.69 MPa - 6 bar (typical CNC) = 87 PSI = 0.6 MPa

I.3.3 Torque

From/To	N*m	kN*m	lb*ft	lb*in	oz*in
1 N*m	1	0.001	0.7376	8.851	141.6
1 kN*m	1000	1	737.6	8851	141,615
1 lb*ft	1.356	0.001356	1	12	192
1 lb*in	0.1130	1.130x10 ⁻⁴	0.08333	1	16
1 oz*in	0.007062	7.062x10 ⁻⁶	0.005208	0.0625	1

I.4 Power and Energy Conversions

I.4.1 Power

From/To	Watt (W)	Kilowatt (kW)	Horsepower (HP)	ft*lb/s
1 W	1	0.001	0.001341	0.7376
1 kW	1000	1	1.341	737.6
1 HP	745.7	0.7457	1	550
1 ft*lb/s	1.356	0.001356	0.001818	1

Spindle Power Example: - 2.2 kW spindle = 2.95 HP - 3 HP spindle = 2.24 kW

I.4.2 Energy and Work

From/To	Joule (J)	Kilojoule (kJ)	Watt-hour (Wh)	Kilowatt-hour (kWh)	BTU
1 J	1	0.001	2.778x10 ⁻⁴	2.778x10 ⁻⁷	9.478x10 ⁻⁴
1 kJ	1000	1	0.2778	2.778x10 ⁻⁴	0.9478
1 Wh	3600	3.6	1	0.001	3.412
1 kWh	3.6x10 ⁶	3600	1000	1	3412
1 BTU	1055	1.055	0.2931	2.931x10 ⁻⁴	1

I.5 Velocity and Acceleration Conversions

I.5.1 Linear Velocity

From/To	mm/s	m/s	m/min	in/s	in/min (IPM)	ft/min
1 mm/s	1	0.001	0.06	0.03937	2.362	0.1969
1 m/s	1000	1	60	39.37	2362	196.9
1 m/min	16.67	0.01667	1	0.6562	39.37	3.281
1 in/s	25.4	0.0254	1.524	1	60	5.0
1 IPM	0.4233	4.233x10 ⁻⁴	0.0254	0.01667	1	0.08333
1 ft/min	5.08	0.00508	0.3048	0.2	12	1

Common CNC Feeds: - 1000 mm/min = 39.37 IPM - 100 IPM = 2540 mm/min = 42.3 mm/s

I.5.2 Rotational Speed

From/To	RPM	rad/s	deg/s
1 RPM	1	0.1047	6.0
1 rad/s	9.549	1	57.30
1 deg/s	0.1667	0.01745	1

I.5.3 Acceleration

From/To	m/s ²	ft/s ²	g (gravity)
1 m/s²	1	3.281	0.102
1 ft/s²	0.3048	1	0.03108
1 g	9.807	32.17	1

CNC Acceleration: 1 m/s² = 60,000 mm/min² (units commonly seen in controller settings)

I.6 Temperature Conversions

From	To Celsius (°C)	To Fahrenheit (°F)	To Kelvin (K)
Celsius	°C	(°C x 9/5) + 32	°C + 273.15
Fahrenheit	(°F - 32) x 5/9	°F	(°F - 32) x 5/9 + 273.15
Kelvin	K - 273.15	(K - 273.15) x 9/5 + 32	K

Common Temperatures: - Water freezing: $0^{\circ}\text{C} = 32^{\circ}\text{F} = 273.15\text{K}$ - Room temperature: $20^{\circ}\text{C} = 68^{\circ}\text{F} = 293\text{K}$ - Water boiling: $100^{\circ}\text{C} = 212^{\circ}\text{F} = 373\text{K}$ - Grease max temp: $150^{\circ}\text{C} = 302^{\circ}\text{F}$

I.7 Material Property Conversions

I.7.1 Density

From/To	kg/m ³	g/cm ³	lb/in ³	lb/ft ³
1 kg/m³	1	0.001	3.613×10^{-5}	0.06243
1 g/cm³	1000	1	0.03613	62.43
1 lb/in³	27,680	27.68	1	1728
1 lb/ft³	16.02	0.01602	5.787×10^{-4}	1

Common Materials: - Steel: $7.85 \text{ g/cm}^3 = 7850 \text{ kg/m}^3 = 0.284 \text{ lb/in}^3$ - Aluminum 6061: $2.70 \text{ g/cm}^3 = 2700 \text{ kg/m}^3 = 0.098 \text{ lb/in}^3$

I.7.2 Hardness (Approximate Conversions)

Rockwell C (HRC) to Brinell (HB) to Tensile Strength (MPa):

HRC	HB	Approx. Tensile (MPa)
20	230	800
30	285	1000
40	370	1300
50	480	1700
60	670	2200

Note: Conversions are approximate, vary by material composition.

I.8 Physical Constants

Constant	Symbol	Value	Units
Gravitational acceleration (Earth)	g	9.80665	m/s ²
Speed of light	c	299,792,458	m/s
Pi	pi	3.14159265359	-
Euler's number	e	2.71828182846	-
Boltzmann constant	k	1.380649×10^{-23}	J/K
Stefan-Boltzmann constant	sigma	5.670374×10^{-8}	W/(m ² *K ⁴)

I.9 Quick Reference Formulas

I.9.1 Cutting Speed to Spindle RPM

$$\text{RPM} = \frac{V_c \times 1000}{\pi \times D}$$

where: - V_c = cutting speed (m/min) - D = cutter diameter (mm)

Example: 100 m/min cutting speed, 10mm endmill

$$\text{RPM} = \frac{100 \times 1000}{\pi \times 10} = 3183 \text{ RPM}$$

I.9.2 Feed Rate Calculation

$$F = f_z \times Z \times \text{RPM}$$

where: - F = feed rate (mm/min) - f_z = chip load per tooth (mm) - Z = number of teeth - RPM = spindle speed

Example: 0.1 mm/tooth, 4-flute, 3000 RPM

$$F = 0.1 \times 4 \times 3000 = 1200 \text{ mm/min}$$

I.9.3 Material Removal Rate (MRR)

$$\text{MRR} = W \times D \times F$$

where: - MRR = material removal rate (mm³/min) - W = width of cut (mm) - D = depth of cut (mm) - F = feed rate (mm/min)

End of Unit Conversions and Constants Appendix